Six-year climatology version (2019 to 2024, same question, season switch, "six years, one shape")

Narration script (2 minutes, Yash then Chau then Yash)

Counted at 140 words per minute (263 words, digits counted as one word) this is about 1:55. Spoken digits add a few seconds, so read briskly; the limit is 2:00 and make_video.sh speeds the audio up to 1.15x if needed. Numbers are generated from eda/facts.json; the team name and title come from team.json. Read it at your natural pace; do not rush to hit the clock. Record in script order. Two-voice version: 3 clips (Yash, Chau, Yash), or 10 clips (one per block) for exact slide sync.

Yash

0:00 to 0:04 (Team name and members, 10 words)

We are Group 19: Chau Nguyen and me, Yash Goel.

0:04 to 0:20 (Dataset and the aspects used, 36 words)

Our dataset is GOES-16's lightning mapper. One file every 5 minutes through six years, 2019 to 2024, averaged to one typical year of 22.5 million sampled flashes on a 1 degree grid by hour and season.

0:20 to 0:28 (Exploratory graph 1, 18 words)

First, where. Land dominates: 81 percent of flashes, hotspots: Lake Maracaibo, northern Colombia, the Amazon, the Gulf coast.

0:28 to 0:35 (Exploratory graph 2, 17 words)

Second, when. Lightning peaks at 4 PM; 1 to 6 PM holds 47 percent of all flashes.

Chau

0:35 to 0:45 (Exploratory graph 3, 22 words)

Third, land against ocean. Land peaks at 4 PM, ocean at 3 AM, and 42 percent of ocean flashes happen at night.

0:45 to 0:59 (Exploratory graph 4, 31 words)

Fourth, each cell's peak hour. 81 percent of land cells peak in the afternoon, 50 percent of ocean cells at night. The Great Plains are the evening patch inside a continent.

0:59 to 1:10 (Exploratory graph 5, 26 words)

Fifth, three regions. The Southeast peaks at 3 PM, the Amazon at 4 PM, but the Plains at 7 PM, 45 percent of it at night.

1:10 to 1:27 (Exploratory graph 6, 38 words)

Sixth, six years, one shape. Land peaked at 4 PM in all six years; the ocean peak drifts between 2 AM and 6 AM on a curve that is nearly flat overnight. A climatology, not one year's weather.

Yash

1:27 to 1:40 (Title of the graph and the question, 31 words)

Our title: When Does Lightning Strike? Day and Night over the Americas. Our question: at what hour does lightning peak over land and ocean, and where does it strike at night?

1:40 to 1:55 (Current progress on the final dynamic graph, 34 words)

Our D3 prototype: play the hour and watch land light up by afternoon, the sea by night. Switch season or clock, click two cells, read the six-year chart underneath. Left: polish and the write-up.

Write-up

What changes from the one-year version

- Data: every complete GOES-16 GLM year, 2019 to 2024 (2,192 days), same 1-in-15 sampling and quality filter. 72 GitHub Actions jobs (one per year and month), 9.5 minutes. 2024 from this run matches the earlier one-year pull exactly.
- The Pacific false-event cell is dropped only in the years it was active (2021 to 2024); in 2019 and 2020 it held ordinary weather (12 and 0 flashes) and is kept.
- The map shows the six-year average: flashes per year in the sample, so the colour scale reads the same as before. 22.5 million good sampled flashes per typical year (the 1-in-15 sample, not the full GLM count).
- Per-year land and ocean curves ship as a few kilobytes of JSON next to the 5 MB counts file.

Design change: a stability chart

- Under the hour chart, "six years, one shape": land (solid) and ocean (dashed) shares by local solar hour, one thin line per year. Twelve lines, two bundles. The reader sees that the rhythm is not one year's weather.
- Land peaked at 4 PM in all six years. The ocean peak drifted between 2 AM and 6 AM on a curve that is nearly flat overnight (the peak hour beats its runner-up by about 0.1 points), so the night share is the robust number.

What six years show

- Land 4 PM, ocean 3 AM; 42 percent of ocean flashes at night; 81 percent of land cells peak in the afternoon, 50 percent of ocean cells at night (both sharper than in 2024 alone, because the noise averages out).
- Autumn is the busiest season (7.1 million per year), then summer (6.7), spring (4.5), winter (4.2).

- 2023 was the busiest year (26.2 million sampled flashes), 2020 the quietest (20.6 million); the hour of the peak did not move.
- Lake Maracaibo peaks at 2 AM; the Great Plains at 7 PM with 45 percent of their lightning at night.

Caveat on the page

- Six years of one satellite at one sampling rate: a climatology of the sample, with GLM's own detection limits built in.

Shared write-up (from the 92-day version)

When Does Lightning Strike? Day and Night over the Americas

Why this question Most people expect thunderstorms in the late afternoon. We want to check that against a satellite that watches the whole hemisphere, and see if it is also true over the ocean. GLM records where and when every flash happens, so we can ask "what hour?" for every 1 by 1 degree cell from Canada to Argentina. A single map of total lightning cannot show timing, so the reader needs to move through the hours.

What we pull right now

- GLM-L2-LCFA files from s3://noaa-goes16/GLM-L2-LCFA/ (NOAA Open Data on AWS). One file every 20 seconds, about 250 KB each, about 50 MB per hour.
- Sample: every 15th file (one per 5 minutes), 1 June to 31 August 2024. That is 92 days, 26,480 files (three days came up a few files short of 288), about 6 GB streamed and never stored. We keep only the counts.
- Open question: the table above says one week in July. The pipeline can do either; one week takes about 20 minutes, the summer takes about 5 hours. We decide after seeing whether one week is enough for the ocean cells, which get few flashes.
- Variables: flash_lat, flash_lon, flash_quality_flag (keep 0 = good), and the hour from the file name. flash_energy and flash_area are read for exploration only.

How we transform it (in addition to the list above)

- Local solar hour = UTC hour + longitude / 15, rounded down.
- Counts per 1 degree cell per hour, once in UTC and once in local solar time. 110 rows by 120 columns by 24 hours, about 1.3 MB of JSON. The page loads it once; there is no server.
- A cell near 50 degrees is only about 64 percent of the area of a cell at the equator, so the map colours flashes per 10,000 km2, not raw counts. The tooltip shows both.
- Land or ocean per cell from the Natural Earth 110m land polygons: a cell is land when at least half of a 4 by 4 grid of points inside it is on land (a single centre test had called the Maracaibo gulf and coastal Cuba sea).
- When comparing cells, each 24 hour curve is divided by its own busiest hour. GLM misses more flashes in daylight and near the edge of its view, so shapes compare fairly and raw totals do not.
- The subtitle says the sampling, period and flash count. The footnote says the normalisation and the detection caveat.

Design choices

- Marks: one square per cell on an equirectangular map. Position = where, colour = flash density in the chosen hour.
- Colour: single hue, light yellow to dark red (YlOrRd), on a square root scale. Densities run from 0 to the thousands, so linear leaves the map empty and log makes 3 flashes look like 300. The legend shows real values. Empty cells are transparent so the coast shows through. The scale is fixed across all 24 hours; otherwise the night map would look as bright as the afternoon one.
- Not dots: millions of flashes as points is a solid blob over Florida and Colombia (overplotting). Not bubbles: area is read worse than colour on a dense grid and circles overlap in the hotspots.
- Hour slider with Play: the core of the design. Play shows the continents light up through the afternoon, go dark around midnight, and the sea hold nearly half of what is left before dawn. Only visible across hours.
- Local solar time / UTC toggle: in UTC the afternoon peak marches west with the sun; in local time it happens everywhere at once. That is why we converted.
- Click a cell: its 24 hour curve appears below the map, scaled to its own peak, next to grey reference curves for all land and all ocean. A second click adds a second cell. A third replaces the oldest.
- Jump buttons: Central Florida (afternoon), Great Plains (night storms), Lake Maracaibo (small hours). Hover shows the count, the density and the share of the cell's day.
- Alternatives we dropped: 24 small static maps (cannot follow one place through the day, no room for curves). Animation with no slider (cannot stop at 3 AM over Nebraska). Land vs ocean lines only (answers the headline but hides where the night lightning is). We kept those lines as the grey reference.

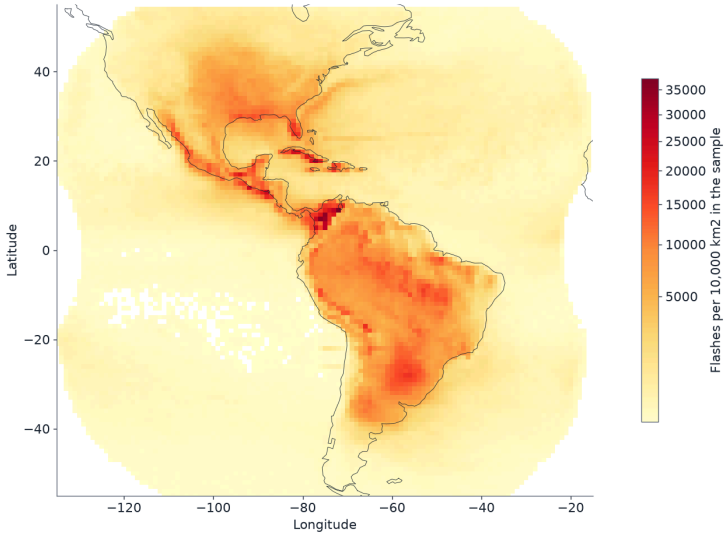
Caveats on the page GLM counts in-cloud plus cloud-to-ground flashes, so we say flashes, not strikes. Detection is weaker by day and at the edge of the view. One summer of one year is a sample, not a climatology.

Development process Work split: [Yash: data pipeline, D3 map and interactions, checkpoint video. Chau: …] People-hours: [about N in total: N data, N D3, N EDA and video, N write-up] What took longest: getting the data out of S3. Files are small but our link gave about 0.5 MB per second no matter how many downloads ran at once, so one day took about 3 minutes and the summer about 5 hours. Opening NetCDF files from several threads crashed (HDF5 is not thread safe); fixed by downloading in parallel and parsing in one thread. In D3, fitting the map to the GLM view took longer than expected: d3 reads polygon winding on the sphere, and a rectangle wound the wrong way selects the rest of the globe.

Exploratory graphs

Where lightning happens: flash density, six-year average

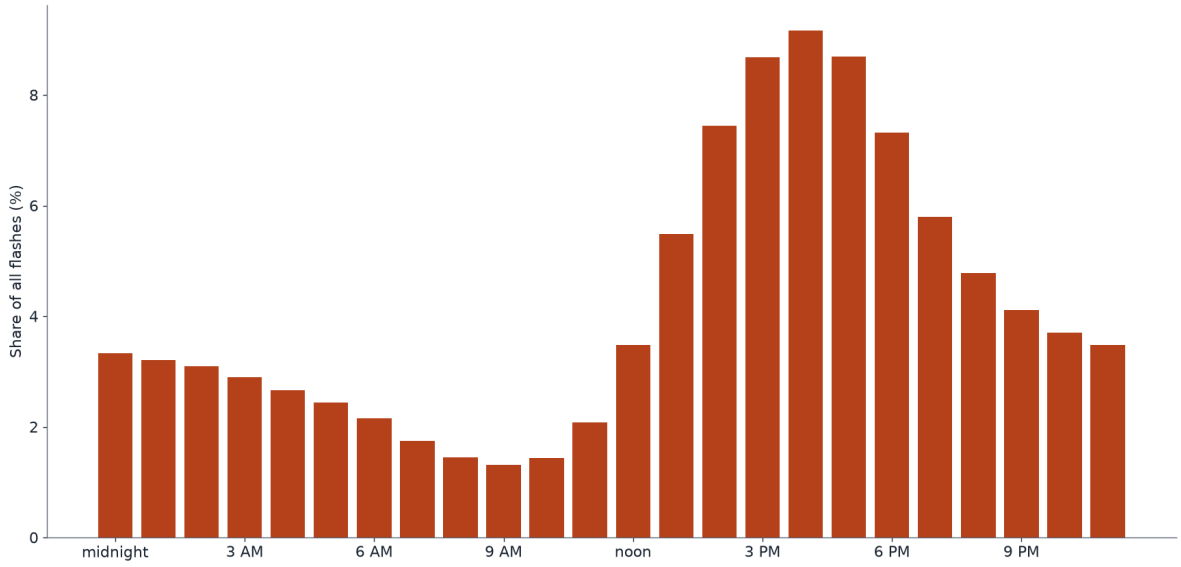
GOES-16 GLM, 2019 to 2024, 2192 days sampled at one 20 s file per 5 min, 22,516,755 flashes per year. Flashes per 10,000 km2 per 1 degree cell per year, square-root colour scale.



g1_map_total

Lightning across the Americas by local solar hour

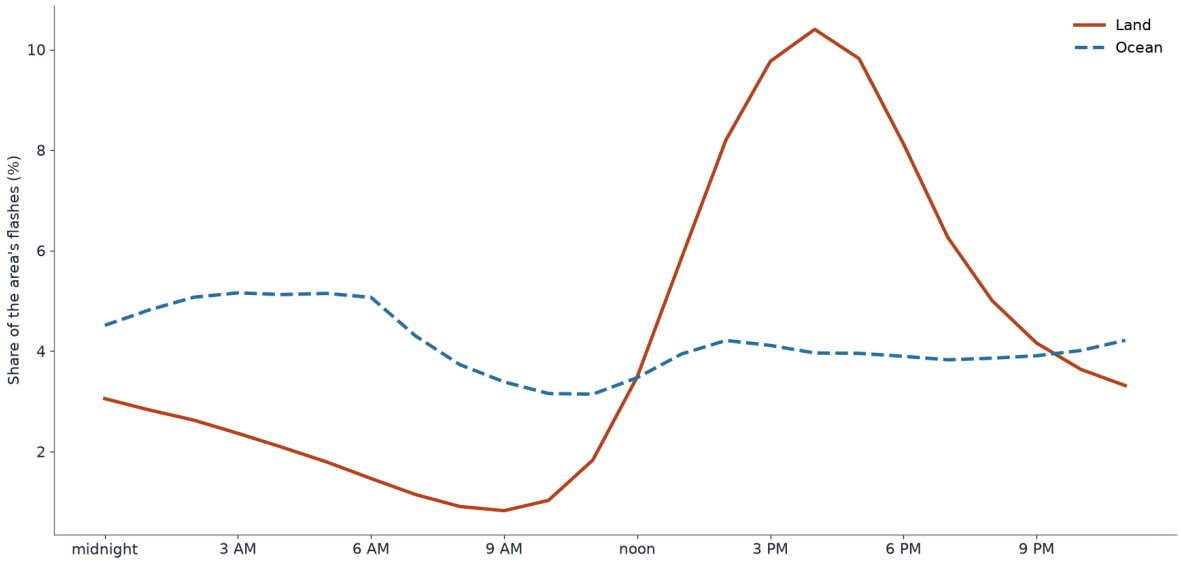
Share of all sampled flashes in each local solar hour (UTC + longitude / 15). Six-year average, all cells.



g2_all_by_hour

Land lightning peaks at 4 PM, ocean lightning at 3 AM

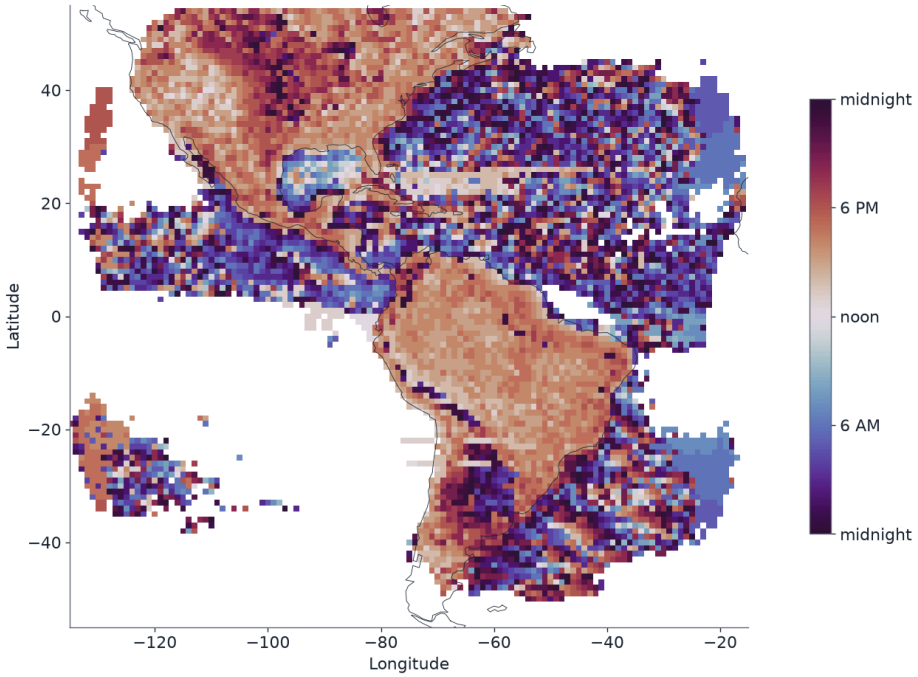
Flashes by local solar hour, all land cells versus all ocean cells, each as a share of its own total.



g3_land_vs_ocean

The hour each place's lightning peaks

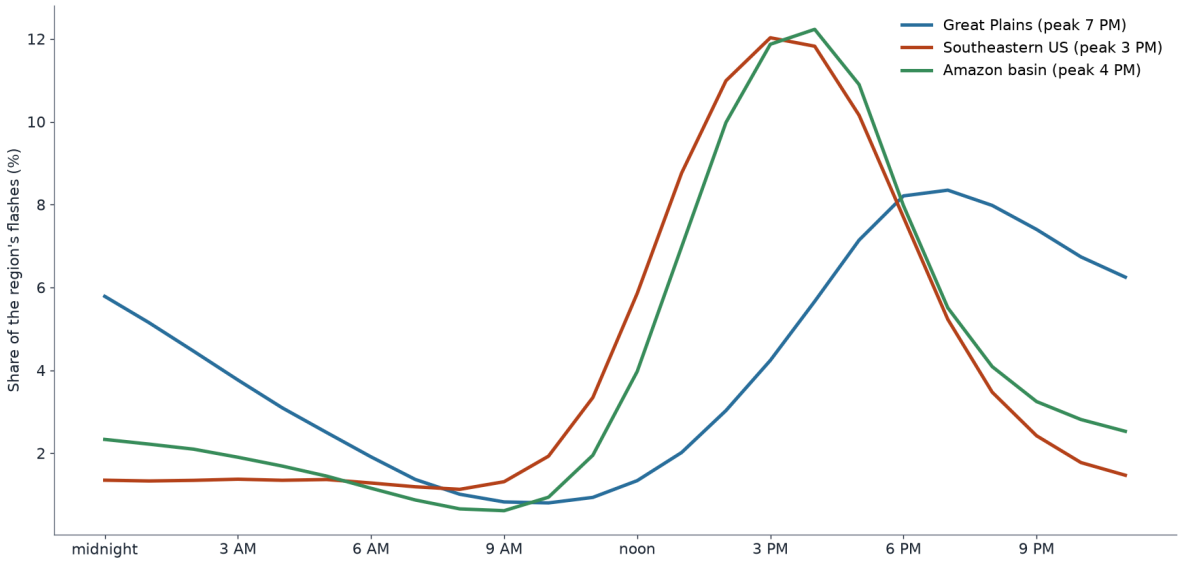
Local solar hour of each cell's busiest hour; cells with under 50 flashes are blank. Circular colour scale, midnight meets midnight.



g4_peak_hour_map

Great Plains peak at 7 PM; Southeast at 3 PM; Amazon at 4 PM

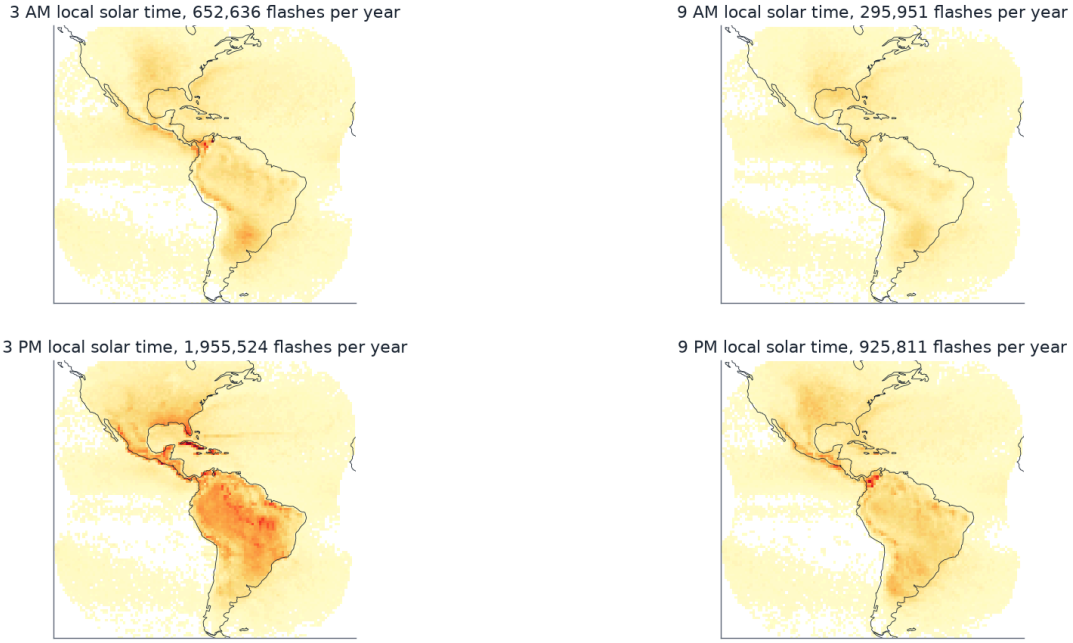
Land flashes by local solar hour, each region as a share of its own daily total.
Night share (9 PM to 6 AM): Plains 45%, Southeast 14%, Amazon 20%.



g5_regions

The same map at four local solar hours

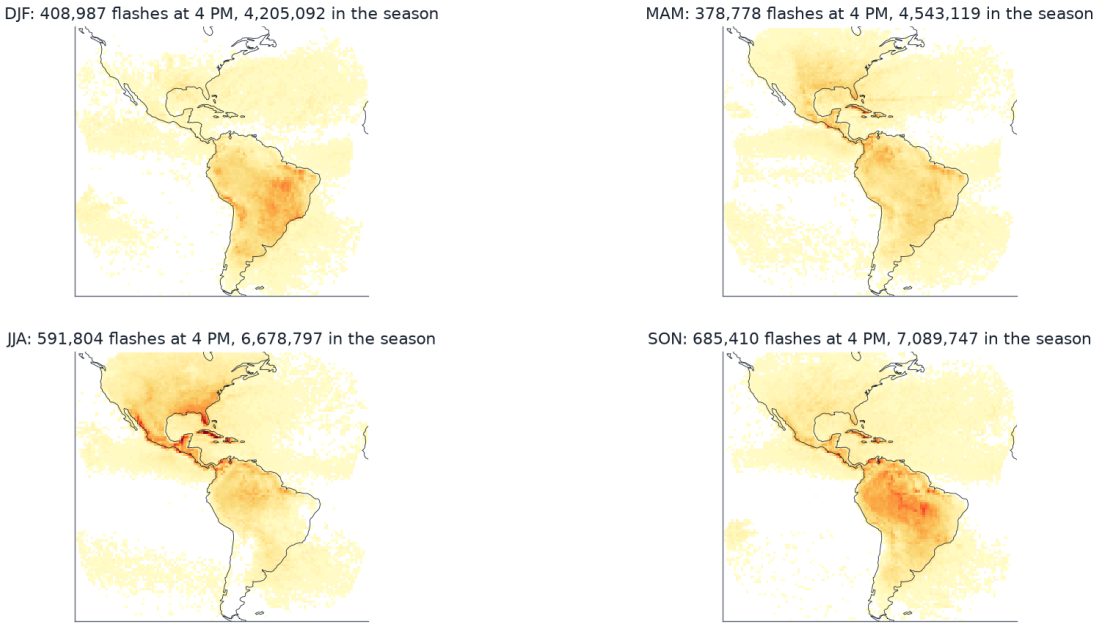
Flashes per cell at 3 AM, 9 AM, 3 PM and 9 PM local solar time. One colour scale for all four panels.



g6_four_hours

The same hour, four seasons: 4 PM local solar time

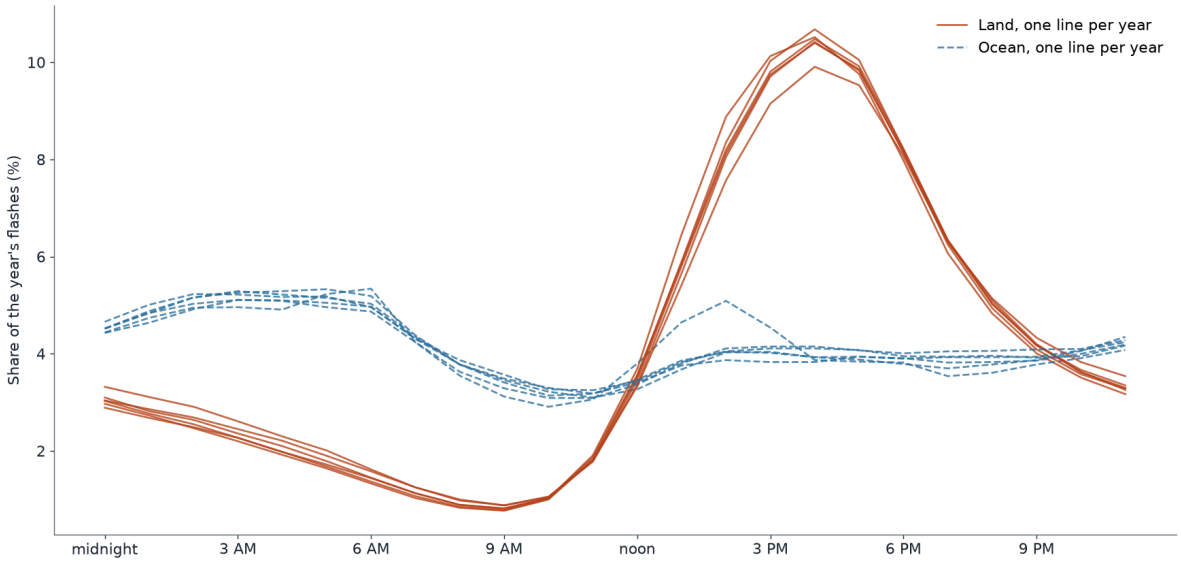
Flashes per cell per year at 4 PM local solar time in each season, six-year average (DJF = Jan, Feb, Dec). One colour scale for all four panels.



g7_four_seasons

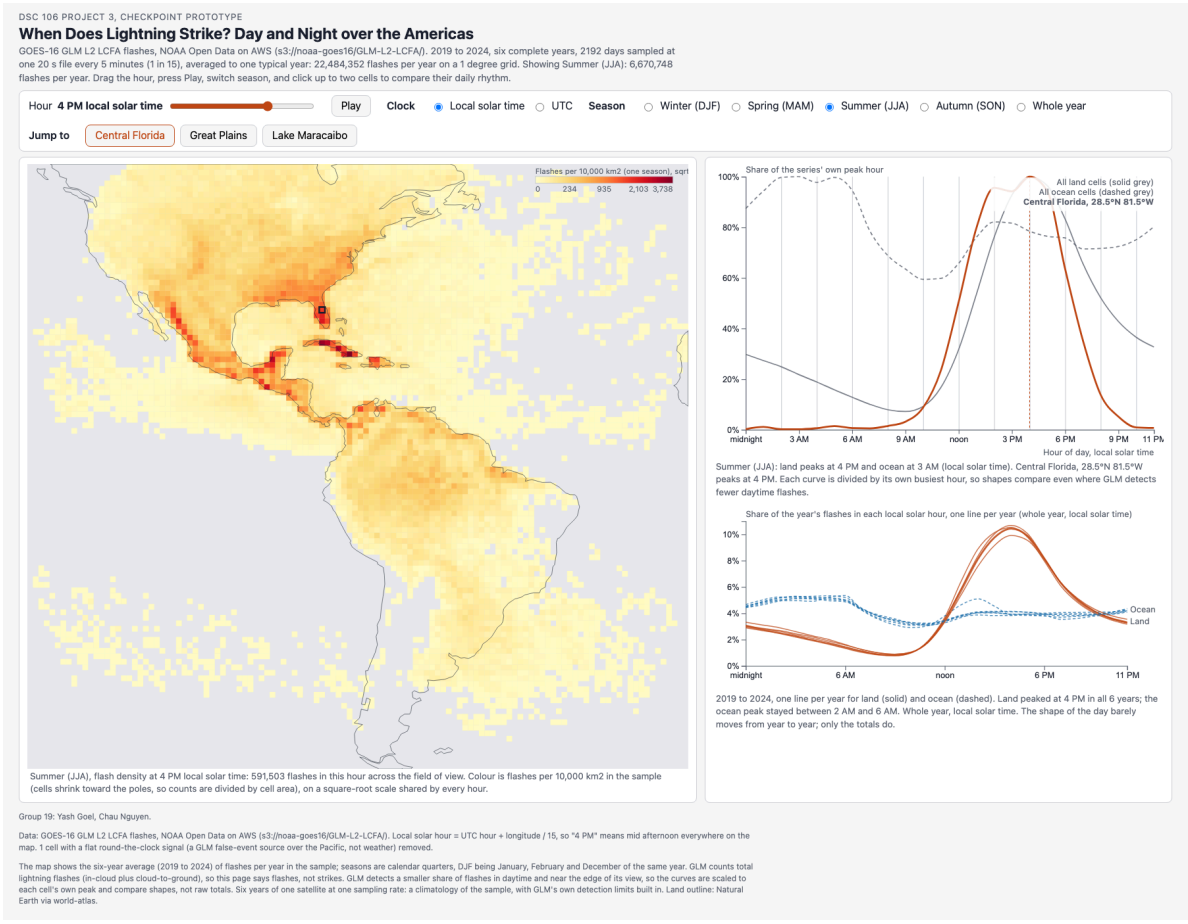
Six years, one shape: the daily rhythm does not move

Land (solid) and ocean (dashed) flashes by local solar hour, one line per year, 2019 to 2024, each as a share of that year's total.

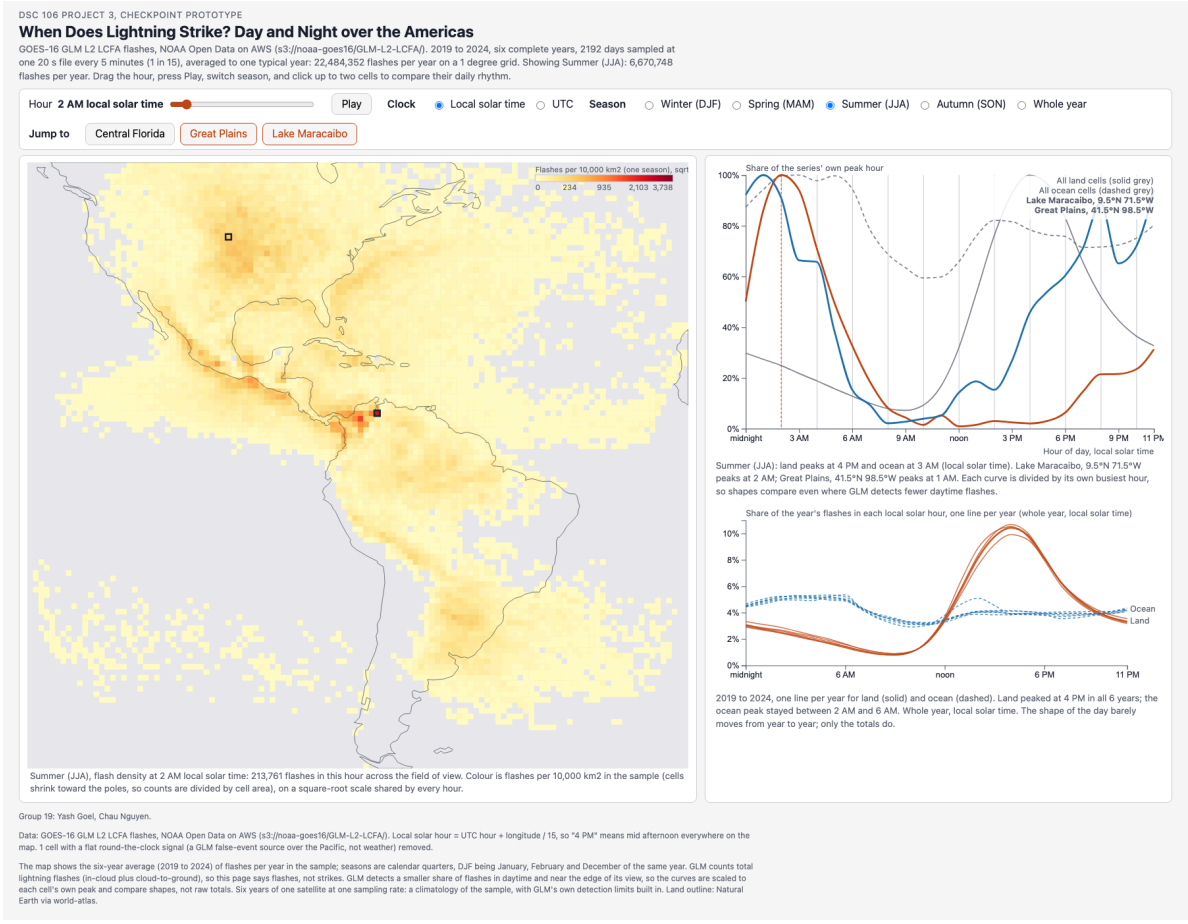


g8_six_years

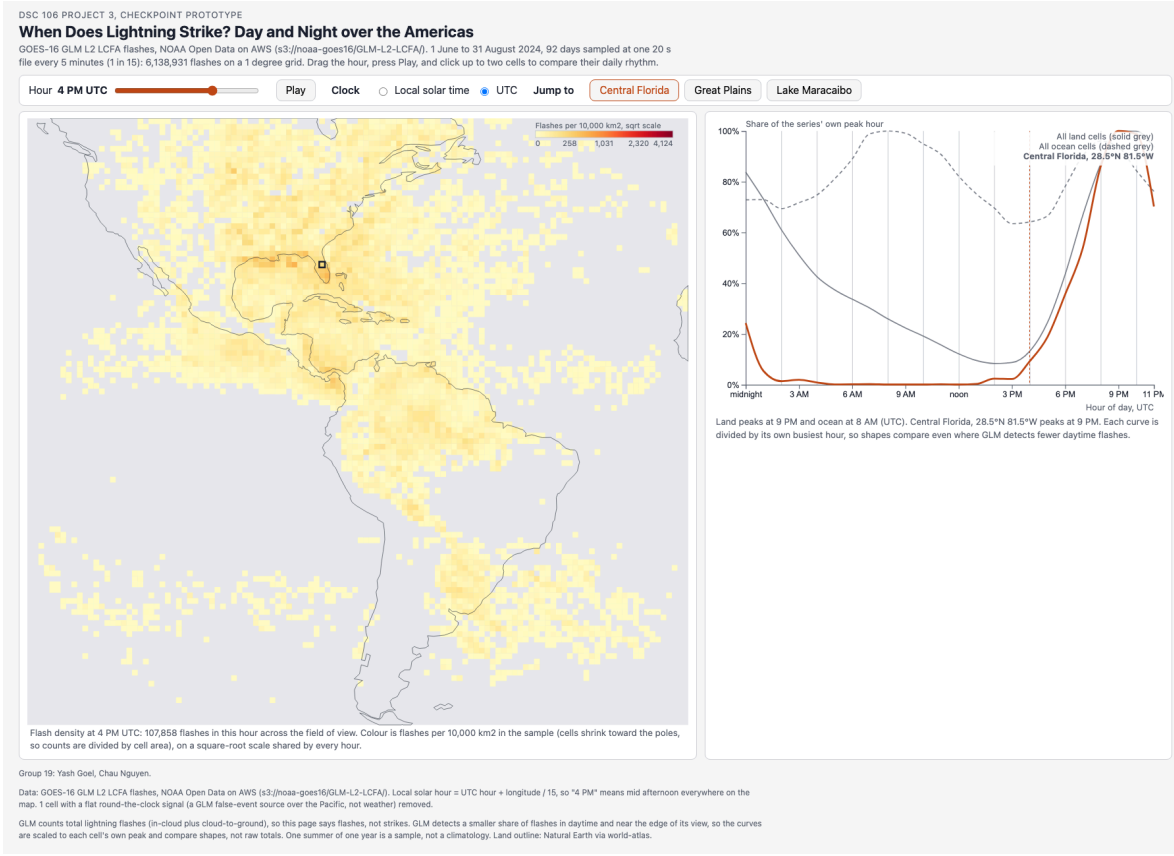
Prototype screenshots



proto_default



proto_night

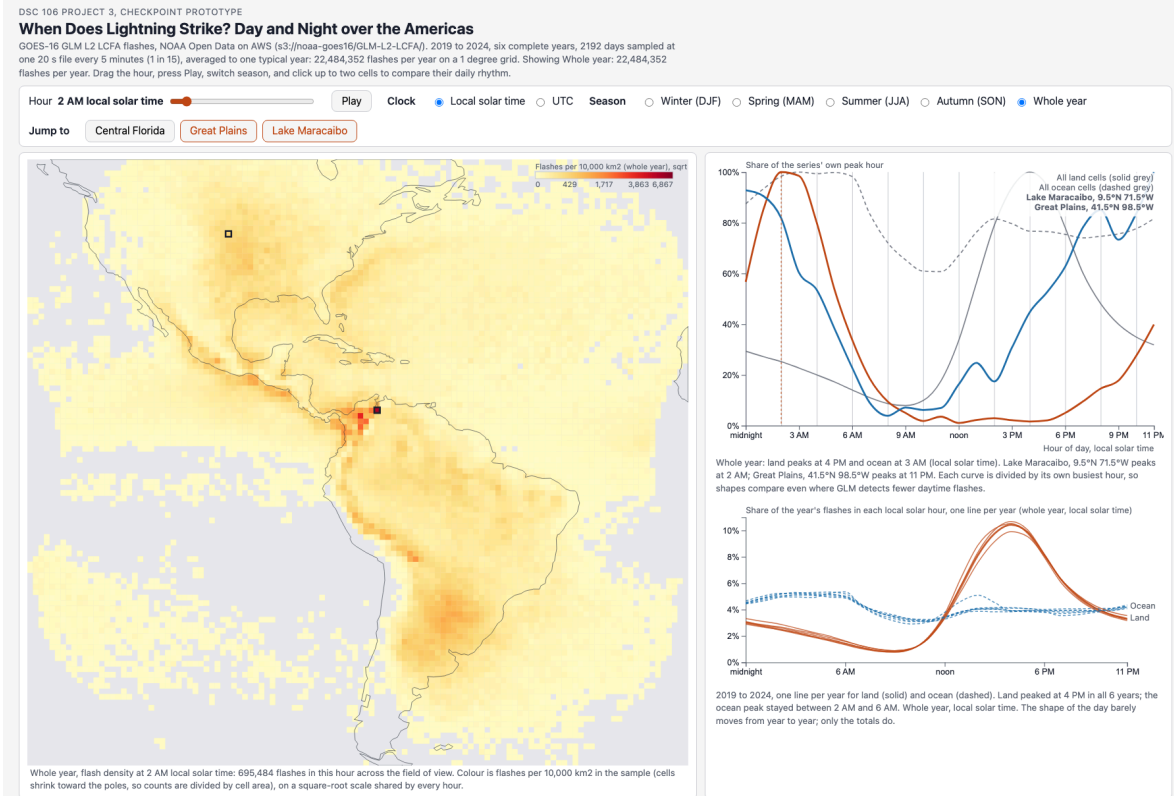


Group 19: Yash Goel, Chau Nguyen.

Data: GOES-16 GLM L2 LCFA flashes, NOAA Open Data on AWS (s3://noaa-goes16/GLM-L2-LCFA/). Local solar hour = UTC hour + longitude / 15, so "4 PM" means mid afternoon everywhere on the map. 1 cell with a flat round-the-clock signal (a GLM false-event source over the Pacific, not weather) removed.

GLM counts total lightning flashes (in-cloud plus cloud-to-ground), so this page says flashes, not strikes. GLM detects a smaller share of flashes in daytime and near the edge of its view, so the curves are scaled to each cell's own peak and compare shapes, not raw totals. One summer of one year is a sample, not a climatology. Land outline: Natural Earth via world-atlas.

proto_utc



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The map shows the six-year average (2019 to 2024) of flashes per year in the sample; seasons are calendar quarters, DJF being January, February and December of the same year. GLM counts total lightning flashes (in-cloud plus cloud-to-ground), so this page says flashes, not strikes. GLM detects a smaller share of flashes in daytime and near the edge of its view, so the curves are scaled to each cell's own peak and compare shapes, not raw totals. Six years of one satellite at one sampling rate: a climatology of the sample, with GLM's own detection limits built in. Land outline: Natural Earth via world-atlas.

proto_year

Video cards

Dataset: GOES-16 GLM

Geostationary Lightning Mapper on NOAA's GOES-16 (GOES-East), Level 2 flash product, NOAA Open Data on AWS

Aspects we use

- Variables: flash_lat, flash_lon, file time stamp, flash_energy, flash_area

- Period: 2019 to 2024, six complete years, 2192 days, one 20 s file every 5 minutes (1 in 15)

- Processing: 22,516,755 flashes per year (six-year mean) on a 1 degree grid by hour and season, UTC and local solar time
- Local solar hour = UTC hour + longitude / 15, so 4 PM means mid afternoon everywhere

Streamed from S3 with boto3 and netCDF4, nothing kept but the counts. About 5 MB of binary counts feeds the D3 prototype.

card_dataset

Our title

When Does Lightning Strike? Day and Night over the Americas

The question we are answering

At what hour of the day does lightning peak over land and over the ocean, and where does it strike at night?

card_question

DSC 106 Project 3: Interactive Visualization, Checkpoint

Team: Group 19

Yash Goel Chau Nguyen

Title: When Does Lightning Strike? Day and Night over the Americas

card_team